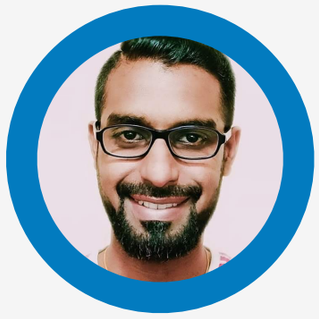
Exercise: Caching Layer



Task 1:

Integrate Redis:

- Cache results of expensive DB queries.
- On cache miss → fetch from DB → store in cache.
- On cache hit → serve instantly.

Concepts: Redis integration, TTL (time-to-live), caching strategy.

```
pip install mysql-connector-python redis
```

Step-1

```
import mysql.connector
import redis
import json
import time

# ----------------------------
# 1. Database Configuration
# ----------------------------
db_config = {
    "host": "localhost",
    "user": "root",
    "password": "password@123",
    "database": "ripad_telemetry",
}

# ----------------------------
# 2. Redis Configuration
# ----------------------------
redis_client = redis.StrictRedis(host="localhost", port=6379, db=0)

```

Step-2

```
20
21 # ----------------------------
22 # 3. Helper Functions
23 # ----------------------------
24
25
26 def get_db_connection():
27     return mysql.connector.connect(**db_config)
28
29
30 def fetch_data_from_db(query):
31     """Fetch result directly from the database."""
32     connection = get_db_connection()
33     cursor = connection.cursor(dictionary=True)
34     cursor.execute(query)
35     result = cursor.fetchall()
36     cursor.close()
37     connection.close()
38     return result
39
```

Step-3

```
40
41 def get_cached_query(query, ttl_seconds=60):
42     """
43     Retrieve data using cache:
44     1. Try Redis cache
45     2. If cache miss → query DB → store in Redis with TTL
46     """
47     # Use query text as cache key (safe for demo; use hashing in production)
48     cache_key = f"cache:{query}"
49
50     # Step 1: Try to get from Redis
51     cached_data = redis_client.get(cache_key)
52     if cached_data:
53         print("Cache hit! Serving from Redis.")
54         return json.loads(cached_data)
55     else:
56         # Step 2: Cache miss → query DB
57         print("Cache miss! Fetching from database...")
58         result = fetch_data_from_db(query)
59
60     # Step 3: Store in Redis with TTL
61     redis_client.setex(cache_key, ttl_seconds, json.dumps(result))
62     print(f"Stored in Redis (TTL={ttl_seconds}s).")
63
64     return result
65
```

Step-4

```
# ------------------------------
# 4. Example Usage
# ------------------------------
if __name__ == "__main__":
    start = time.time()
    query = "SELECT id,sensor_id,asset_id,temperature,pressure,vibration,metrics
             from ripad_telemetry.telemetry_data"
    data = get_cached_query(query, ttl_seconds=30)
    print(f"Fetched {len(data)} rows in {round(time.time() - start, 3)}s")

    print("\nRunning the same query again (should hit cache)...\n")
    start = time.time()
    data = get_cached_query(query, ttl_seconds=30)
    print(f"Fetched {len(data)} rows in {round(time.time() - start, 3)}s")
```